

CONSTRUCTION OS

PRD_Work_Progress_Module

Work Progress, Site Operations and Construction Control

Product definition

The construction execution source of truth for work packages, baselines, planned and actual progress, measured quantities, daily site records, crews, materials, equipment, delays, constraints, evidence, subcontractor claims, progress certification and reliable completion forecasting. The module replaces subjective percentage reporting with controlled, evidence-backed and auditable progress records.

Field	Value
Document type	Full Product Requirements Document
Status	Implementation-ready functional and technical specification
Version	1.0
Date	5 August 2026
Priority	Critical construction operations module
Audience	Product, UX/UI, Frontend, Backend, Data, QA, Security and DevOps teams
Core outcome	A trusted daily operating system that makes physical progress, delay, productivity and completion risk visible before they become financial failure
Approval architecture	Local verification states and future integration hooks only; the platform-wide Approvals and Workflow Engine remains intentionally deferred

Design direction: premium, field-ready, measurable, immutable and construction-specific.

1. Document Control

Field	Value
Document	PRD_Work_Progress_Module - Work Progress, Site Operations and Construction Control
Primary routes	/work-progress, /projects/{projectId}/work-progress, /work-packages/{id}, /daily-reports, /progress-certification
Primary pages	Work Progress Dashboard, Work Packages, Work Package Page, Construction Schedule, Lookahead Planning, Daily Site Reports, Delays and Constraints, Progress Photos, Subcontractor Progress, Measurement and Certification, Reports and Settings
Related PRDs	Projects and Project Page, Tasks, Documents, Teams, Contracts, Finance, Inventory, Assets, Procurement, Units and future QA/QC, HSE, Calendar, Notifications, BIM/3D and Approvals
Current access model	All authenticated users may use the complete feature set in the current implementation phase. Every action must still call the common authorization interface so later role, company, project, contractor, record and field restrictions require no redesign.
Platform Configuration	Mandatory at module, page, section, widget, field group, calculation method, status, workflow checkpoint, report and action level
Core source-of-truth rule	Work Progress owns physical construction execution records, work packages, accepted progress measurements, site diaries, delay and constraint records, operational forecasts and progress snapshots. Connected modules remain authoritative for their own contracts, costs, stock, assets, files, people, inspections, incidents and decisions.
Data integrity rule	Official progress is derived from accepted measurement records against an active baseline. Issued baselines, submitted daily reports and accepted progress entries are never overwritten; corrections use revisions or reversal entries.
External integrations	Not required in initial implementation. Import/export and integration contracts remain available for Primavera P6, Microsoft Project, BIM/IFC, drone, weather and field-device integrations later.
Approval dependency	This PRD defines verification checkpoints and decision payloads but does not define organization-wide routing, delegation, escalation or approval policy.

2. Table of Contents

- 3. Executive Summary
- 4. Why Construction Companies Fail Without This Module
- 5. Product Vision, Goals and Non-Goals
- 6. Architectural Ownership and Domain Boundaries
- 7. Users, Personas and Responsibility Model
- 8. Information Architecture and Navigation
- 9. Construction Data Hierarchies
- 10. Work Package Architecture
- 11. Baselines, Schedule Versions and Change Control
- 12. Work Progress Dashboard
- 13. Work Packages Module Page
- 14. Work Package Page
- 15. Construction Schedule and Lookahead Planning
- 16. Progress Measurement Engine
- 17. Progress Rollups, Weighting and Formula Registry
- 18. Daily Site Reports and Site Diary
- 19. Field Capture, Mobile and Offline Operation
- 20. Workforce, Crews and Productivity
- 21. Materials, Deliveries and Consumption
- 22. Equipment and Asset Utilization
- 23. Subcontractor Progress and Commercial Measurement
- 24. Delays, Constraints, Risks and Recovery Planning
- 25. Evidence, Progress Photos and Location Capture

- 26. Quality, HSE and Readiness Gates
- 27. Progress Claims, Measurement Sheets and Certification
- 28. Cost, Finance, Contracts and Change Integration
- 29. Forecasting, Completion Confidence and Data Quality
- 30. Search, Filters, Saved Views and Reporting
- 31. Notifications, Calendar and Collaboration Hooks
- 32. Documents and Record Folder Architecture
- 33. Responsive UX, Premium Motion and Accessibility
- 34. Platform Configuration and Localization
- 35. Functional Data Model
- 36. Lifecycle and Status Models
- 37. API Contract
- 38. Backend Services, Events and Background Jobs
- 39. Frontend Architecture
- 40. Offline Synchronization and Conflict Resolution
- 41. Validation, Concurrency, Security and Audit
- 42. Performance, Scalability and Observability
- 43. Testing Strategy and Quality Gates
- 44. Import, Migration and Rollout
- 45. Implementation Phases
- 46. Acceptance Criteria
- 47. Page and Flow Registry
- 48. Future Roadmap
- 49. Final Architectural Rule

3. Executive Summary

The Work Progress Module is the operational control center for physical construction execution. It converts project scope, schedule, BOQ quantities, locations, contractors, crews, materials, equipment and field evidence into one auditable account of what was planned, what was physically completed, what was verified, what is blocked and what is likely to finish late.

The module must eliminate the common failure pattern in which every department reports a different progress percentage. One project may currently show 70 percent in a presentation, 62 percent in a contractor claim, 58 percent in the schedule and 54 percent in measured quantities. Construction OS must preserve each context separately, identify which value is official for each purpose and expose the calculation behind every rollup.

Core product statement

Tasks coordinate people. Work Progress records physical execution. Finance records cost. Contracts define obligations. QA/QC validates quality. HSE governs safety. Work Progress connects these sources without replacing them and produces the trusted operational picture of project delivery.

- **Evidence-backed progress.** Every official progress increase must be attributable to an accepted quantity, milestone, unit, activity or controlled manual measurement and, where configured, supporting evidence.
- **Baseline discipline.** Planned progress is calculated only against an identified active baseline. Rebaselining never erases the original plan or historical variance.
- **Fresh field data.** Daily reports, photos, measured quantities, crews and blockers must be captured close to the time and location of work, including offline operation.
- **No double counting.** The measurement engine prevents the same quantity, location or milestone from contributing twice to project progress.
- **Commercial separation.** Contractor claimed progress, site-measured progress, accepted physical progress and financially certified progress are separate states.
- **Actionable forecasting.** The module uses current production rates, remaining quantities, constraints and schedule dependencies to explain likely completion dates and recovery requirements.
- **Immutable history.** Accepted updates, submitted site reports, active baseline snapshots and certification records remain historically reproducible.

4. Why Construction Companies Fail Without This Module

Construction failure is often visible in site data weeks or months before it appears in cash flow, claims or final delay. The platform must detect and structure these signals early.

Failure mode	Operational symptom	Required platform control
Subjective percentage reporting	Progress is entered as an unsupported number and changes depending on who reports it.	Configurable measurement methods, accepted quantities, milestone evidence and visible calculation lineage.
No active baseline	Planned progress changes silently and delay disappears after schedule edits.	Immutable baseline versions, active-baseline identity, approved rebaseline hooks and variance against original and current baseline.
Late or missing site reporting	Management sees problems only after weekly or monthly meetings.	Daily reporting deadlines, stale-update warnings, mobile/offline capture and exception dashboards.
Double-counted work	The same installed work appears in multiple packages, claims or locations.	Unique measurement dimensions, overlap validation, quantity caps and reversal-based corrections.
Progress disconnected from scope	The project is 60 percent complete but nobody can explain which quantities remain.	WBS, LBS, BOQ and work-package mappings with drill-down to remaining quantity and location.
Progress disconnected from schedule	A high overall percentage hides late critical activities.	Dependency-aware schedule, critical/near-critical views, float, forecast dates and milestone risk.
Progress disconnected from cost	Cash has been spent but little physical value was produced.	Finance-linked actual cost and optional earned-value indicators without duplicating accounting.
Contractor overclaiming	Claimed quantities exceed measured or accepted work.	Claim, measured, accepted, certified and paid values stored separately with variance reasons.
Hidden constraints	Permits, drawings, materials or access problems remain in messages and meetings.	Structured constraint register with owner, required-by date, impact, linked evidence and release history.

Failure mode	Operational symptom	Required platform control
Poor evidence	Photos exist but cannot be connected to a date, location or work package.	Controlled photo metadata, location links, annotations, duplicate detection and evidence coverage checks.
Retrospective manipulation	Old reports or progress values are edited after disputes arise.	Record locking, immutable activity, revision history, timestamps and audit exports.
False recovery plans	Recovery is promised without quantified resources or production rates.	Recovery scenarios with added crews, shifts, equipment, sequence changes, cost impact and forecast result.
Unreliable rollups	A small expensive package distorts physical progress or a large scope has no weight.	Explicit weighting basis, weight total validation, formula registry and reproducible snapshots.
Offline data loss	Site users work without connectivity and later re-enter incomplete information.	Offline queue, client-generated IDs, resumable uploads, idempotent APIs and conflict handling.
No ownership	Every delay is known but no person is responsible for removing it.	Required accountable owner, escalation due date and aging status on constraints and delays.

WP-001 The system must never display one unexplained project progress number. Every percentage must identify its type, baseline, effective date, measurement method and drill-down source.

WP-002 The system must distinguish at minimum: planned progress, submitted progress, accepted physical progress, contractor claimed progress, certified progress and financial payment progress.

WP-003 Changing a baseline, scope quantity, work-package weight or measurement method must never rewrite previously published progress snapshots.

5. Product Vision, Goals and Non-Goals

5.1 Product Vision

- Create one trusted construction execution record across all companies, projects, sites, structures, floors, zones, units, disciplines and contractors.
- Make physical progress measurable from scope and quantity rather than presentation-driven estimates.
- Give site teams a fast daily tool and management a reliable portfolio-level control system from the same records.
- Connect schedule, cost, contract, procurement, inventory, assets, workforce, documents, QA/QC and HSE without duplicating their authoritative data.
- Provide full traceability from executive progress charts to the individual accepted measurement and evidence item.
- Predict late completion from production, remaining scope, dependencies and constraints before contractual milestones fail.
- Support a premium web experience across desktop, tablet and mobile, with offline-first field workflows.

5.2 Product Goals

- Know exactly what work is planned, ready, active, blocked, complete and accepted.
- Know where work occurred and which contractor, crew, shift, material and equipment contributed.
- Compare planned, actual and forecast progress by project, structure, floor, zone, discipline, package, contractor and reporting period.
- Capture daily site conditions, labor, equipment, deliveries, completed quantities, issues, visitors and tomorrow plan.
- Prevent progress from exceeding approved scope or being counted in overlapping records.
- Detect stale data, missing evidence, missing quantities, blocked critical work and deteriorating production rates.
- Support progress claims and certification without replacing Contracts or Finance.
- Produce daily, weekly, monthly, client, contractor and executive reports from the same immutable data.
- Scale from a small renovation to multi-tower, multi-company projects with millions of progress lines and photos.

5.3 Non-Goals

- Personal task collaboration, reminders and participant-based discussions; Tasks remains authoritative.
- Legal contract authoring, terms, claims adjudication or extension-of-time decisions; Contracts and future Approvals remain authoritative.
- Accounting entries, invoices, payments, payroll or statutory finance; Finance remains authoritative.
- Warehouse balances, serial/batch stock ledger or inventory valuation; Inventory remains authoritative.

- Asset identity, maintenance, depreciation or ownership; Assets remains authoritative.
- Full QA/QC inspection, NCR and handover lifecycle in this release; the module consumes quality states and links records.
- Full HSE incident, permit and risk-assessment lifecycle in this release; the module consumes safety blocks and links records.
- Organization-wide approvals, routing, delegation and escalations; only future-ready checkpoints are included.
- Native mobile applications, live drone processing, BIM authoring or AI prediction in the first release.

6. Architectural Ownership and Domain Boundaries

Domain	System of record	Work Progress behavior
Project identity and hierarchy	Projects	References project, phase, structure, floor, zone, unit and project status. Does not duplicate project master data.
Work packages and physical execution	Work Progress	Owns package scope, readiness, progress method, field updates, accepted measurements, delays, constraints, snapshots and operational forecasts.
Construction schedule and baseline	Work Progress	Owns operational activities, dependencies, calendars, schedule versions and progress baseline used by this module. May import external schedules later.
Tasks and action items	Tasks	Creates or links tasks for corrective work, missing data, constraint removal and follow-up. Task status does not automatically equal physical progress.
Files, drawings and media	Documents	All controlled files remain Document records. Progress stores references, evidence metadata and optional media renditions.
Contract obligations and claims	Contracts	Reads milestones, scope, parties and contractual dates. Publishes progress facts and linked claim evidence; does not alter contract terms.
Finance and project cost	Finance	Reads budget, commitment and actual-cost summaries. Publishes accepted quantity/progress context and certification-ready values.
Products, stock and consumption	Inventory	Reads products and linked stock issues/returns. Inventory owns quantity movement and valuation; Work Progress owns the operational allocation context.
Assets and equipment	Assets	Reads asset identity, assignment, meter and utilization records. Work Progress stores work-package usage links and operational context.
Purchase orders and deliveries	Procurement	Reads expected delivery and procurement package status; links delays and receipts to affected work.
Employees and workforce	HR / User Profile / Teams	Reads worker identity, employer, role, qualification and attendance where authorized. Work Progress owns crew allocation and productive-hours context.
Quality inspections and defects	Future QA/QC	Reads hold points, inspection result, NCR and handover status. Quality acceptance may gate progress acceptance.
Safety permits and incidents	Future HSE	Reads permit, safety stop and incident state. HSE blocks may stop work-package readiness or progress acceptance.
Meetings and calendar	Future Meetings / Calendar	Publishes deadlines, lookahead events and review meetings. Meetings owns agendas, minutes and decisions.
Decisions and approvals	Future Approvals Engine	Work Progress emits structured requests for baseline activation, rebaseline, manual overrides, certification and disputed measurement decisions.

WP-004	Cross-module records must be linked by immutable IDs and correlation IDs. No module may copy a linked record into an editable local duplicate.
WP-005	A Task completion may trigger a suggested progress update but must never increase official physical progress automatically without a valid measurement record.
WP-006	Finance payment status must not be used as a substitute for physical progress, and physical progress must not be used as proof of payment.

7. Users, Personas and Responsibility Model

Persona	Responsibilities in this module
Project Director	Portfolio/project oversight, baseline awareness, milestone risk, recovery decisions and executive reporting.
Project Manager	Own package delivery, schedule coordination, constraints, forecasts, contractor performance and project reporting.
Construction Manager	Coordinate active works, sequences, interfaces, readiness and recovery operations.
Site Engineer	Capture and verify quantities, daily execution, evidence, locations and technical blockers.
Planning Engineer	Maintain schedules, baselines, dependencies, lookaheads, progress curves, critical path and forecast dates.
Quantity Surveyor / Commercial Manager	Review BOQ mappings, contractor claims, measured quantities, certification and commercial variance.
Site Supervisor / Foreman	Record crew deployment, completed work, production rates, issues and next-shift plan.
Subcontractor Representative	Submit permitted progress claims, daily records and evidence limited to assigned packages in a later restricted-access phase.
Document Controller	Ensure current drawings, method statements and records are linked without duplicating Documents data.
Warehouse / Logistics User	Link deliveries, material issues and shortages to work packages and daily reports.
Asset / Equipment Coordinator	Link equipment assignments, operating hours, downtime and availability constraints.
QA/QC User	Provide inspection, hold-point, defect and acceptance status that may gate accepted progress.
HSE User	Provide permit, safety-stop and incident status that may gate readiness and execution.
Finance User	Read accepted physical progress and certification context; provide actual cost and payment states.
Client / Consultant Reviewer	Read authorized progress, evidence, reports and measurement/certification records in a future controlled-access phase.
Auditor	Reconstruct baselines, changes, accepted updates, certifications, evidence and actor history.

7.1 Responsibility Model

- **Accountable owner.** Every work package, delay, constraint, recovery plan, daily report and measurement sheet must have one accountable owner.
- **Responsible contributors.** Multiple users, teams, contractors or crews may contribute without replacing the accountable owner.
- **Verifier.** A progress record may require a designated verifier separate from its submitter.
- **Commercial reviewer.** Claimed and certified progress may require a quantity surveyor or commercial reviewer.
- **Observer / watcher.** Authorized users may follow records without gaining edit rights.
- **Role snapshots.** The actor role, company and project relationship at the time of each action must be retained for audit even if employment later changes.

8. Information Architecture and Navigation

Navigation area	Pages / workspaces
Overview	Global Work Progress Dashboard, project selector, portfolio exceptions and recent activity
Project execution	Project Work Progress Dashboard, Work Packages, Work Package Page, Location/Discipline views
Planning	Construction Schedule, Baselines, Milestones, Lookahead Planning and Readiness
Field operations	Daily Site Reports, Shift Reports, Quick Progress Capture, Crew and Equipment Logs
Control	Delays, Constraints, Recovery Plans, Data Quality and Stale Updates
Evidence	Progress Photos, Videos, Drawings, Annotations and Evidence Coverage
Commercial measurement	Subcontractor Claims, Measurement Sheets, Certification and Disputes
Analytics	S-curves, production rates, contractor comparison, remaining quantities, forecast and reports
Administration	Measurement Methods, Statuses, Calendars, Units, Code Formats, Templates and Platform Configuration

8.1 Entry Points

Entry point	Behavior
Project Page -> In-Progress Works	Open project Work Packages filtered to active, delayed, blocked or recently updated packages.
Project Page -> Progress indicator	Open project dashboard with the calculation drill-down for the displayed progress type.
Global sidebar -> Work Progress	Open portfolio dashboard across authorized projects and companies.
Task / Contract / Procurement / Inventory / Asset record	Open linked work package, delay, daily report, progress update or measurement sheet.
Global Search	Open a specific package, report, delay, constraint, update or certification record.
3D / BIM viewer later	Open the linked package or progress capture view with model element and location context.
Mobile quick action	Start a daily report, quantity update, photo capture, blocker or equipment log for a recently used project.

WP-007

Project Page In-Progress Works and Tasks must remain visibly separate concepts and use different status, progress and data semantics.

WP-008

Every project-level progress summary must support drill-down to structure, location, discipline, work package, activity and accepted measurement where those dimensions exist.

9. Construction Data Hierarchies

Construction control requires multiple independent hierarchies. They must be linked but never collapsed into one overloaded tree.

Hierarchy	Purpose	Examples / rules
Project Breakdown	Legal and operational project context	Project -> Phase -> Structure/Tower -> Floor/Level -> Zone/Area -> Unit. Owned by Projects; nullable levels allowed.

Hierarchy	Purpose	Examples / rules
Work Breakdown Structure (WBS)	Organize deliverable scope	Project -> Major deliverable -> Work package -> Activity. Configurable codes and parent-child rollups.
Location Breakdown Structure (LBS)	Identify where work is executed	Site, building, floor, room, facade elevation, grid, chainage, geographic segment or custom location.
Cost Breakdown Structure (CBS)	Connect execution to cost control	Cost center, cost code, budget line, account mapping and contract line. Finance remains authoritative.
BOQ / Quantity Structure	Measure scope	BOQ section, item, unit, approved quantity, rate and variation links. Contract/Finance source linkage where applicable.
Organization Structure	Identify responsibility	Owning company, department, team, contractor, subcontractor, crew and responsible user.
Schedule Structure	Plan sequence and time	Calendar, activity, milestone, dependency, constraint date, float and baseline version.
Document / Model Structure	Connect information	Drawing package, document revision, model element, system, tag or spatial object.

9.1 Mapping Rules

- A work package may map to multiple locations, BOQ lines, schedule activities, contracts, documents and cost codes.
- A schedule activity may contribute to one or more work packages, but each progress contribution must declare its allocation to prevent double counting.
- A BOQ line may be divided across work packages or locations using controlled allocation percentages or quantities whose total cannot exceed the approved scope.
- Project structures and units remain referenced by ID. Deleting or archiving a hierarchy node must preserve historical progress links.
- Custom dimensions may be configured but must not replace required project, package, date, status and measurement fields.
- Every rollup must disclose the hierarchy and weighting rule used.

WP-009	The sum of BOQ quantity allocations across active work packages must not exceed the current approved BOQ quantity unless an authorized overrun or pending variation state is explicitly recorded.
WP-010	The system must permit projects without buildings, floors or units. Null hierarchy levels must not create artificial placeholder records.

10. Work Package Architecture

A Work Package is the primary execution record for a defined scope of physical work. It is not a task, contract, BOQ item or schedule activity, although it may link to all four.

10.1 Work Package Examples

- Tower A reinforced concrete frame
- North elevation facade installation
- Level 4 electrical first fix
- Apartment Type B flooring installation
- Site-wide drainage network
- Road Section CH 0+000 to CH 1+500
- Mechanical plant room commissioning
- External landscaping Zone 3
- Demolition and enabling works
- Custom non-building infrastructure package

10.2 Required Work Package Fields

Field group	Required / supported fields
Identity	Permanent ID, package code, name, description, discipline, category, package type, status and priority
Ownership	Tenant, owning company, project, accountable manager, responsible team, contractor/subcontractor and commercial owner
Scope	WBS parent, locations, BOQ lines, quantities, unit, scope inclusions, exclusions, deliverables and completion definition
Time	Planned start/finish, current baseline dates, forecast dates, actual dates, calendar, schedule links and milestones
Progress	Measurement method, weight, submitted percent, accepted percent, certified percent, last accepted update and data freshness
Readiness	Required documents, drawings, permits, materials, equipment, access, predecessor completion, quality hold points and safety gates
Commercial	Contract/subcontract, budget/cost codes, rates, claim method, retention context and variation links
Evidence	Photos, videos, measurement sheets, inspection links, documents, model elements and annotations
Control	Constraints, delays, risks, recovery plan, change history, notes, activity timeline and archive reason

10.3 Work Package Lifecycle

Draft	Planned	Ready	In Progress	Suspended	Substantially Complete	Complete	Closed
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Cancelled and Archived are controlled terminal states. A reopened package creates a lifecycle event and retains prior completion history.

State	Meaning	Minimum transition rules
Draft	Scope is being defined and does not contribute to official progress.	Project and identity required.
Planned	Scope, baseline dates and measurement method are defined.	Approved scope quantity/weight and active baseline linkage required.
Ready	All configured readiness gates are satisfied.	No blocking predecessor, permit, document, material, equipment, QA/QC or HSE gate.
In Progress	Physical execution has started.	Actual start and responsible party required.
Suspended	Execution intentionally stopped.	Reason, start time, owner and forecast impact required.
Substantially Complete	Primary scope is operationally complete; minor residual work remains.	Residual list and forecast closeout date required.
Complete	Configured physical completion definition is met.	Accepted progress must equal completion threshold and required evidence/quality gates must pass.
Closed	Commercial, document and operational closeout is complete.	No unresolved mandatory closeout item.
Cancelled	Package will not be executed.	Reason, scope treatment and rollup impact required.
Archived	Read-only historical record.	Only after terminal state and retention checks.

WP-011

A work package cannot become Ready while a configured blocking gate is unresolved. Authorized overrides must be explicit, time-bound and audited.

WP-012

A work package cannot become Complete solely because the planned finish date has passed or a Task was marked complete.

WP-013

Negative progress corrections must use a reversal or correction record and must never silently lower an accepted historical entry.

11. Baselines, Schedule Versions and Change Control

The module must preserve the difference between the original promise, the current approved plan and the latest operational forecast.

Schedule / scope concept	Definition
Original Baseline	First formally activated scope, weights, quantities, dates and planned curve. Immutable after activation.
Current Approved Baseline	The active approved plan used for primary planned-versus-actual reporting. A project has exactly one active version per configured scope.
Interim Working Schedule	Editable planning version used before activation. Does not alter official planned progress.
Rebaseline	A new approved baseline version created due to authorized change. It supersedes but never deletes the prior version.
Forecast	Best current prediction based on remaining scope, production and constraints. It may change without becoming a baseline.
What-if Scenario	Private or shared simulation that never affects official progress until explicitly promoted through a future decision checkpoint.
Progress Snapshot	Immutable time-stamped result containing planned, actual, forecast, quantities, weights and calculation version for later reproduction.

11.1 Baseline Content

- Baseline ID and version
- Scope effective date
- Prepared by and activation timestamp
- Included work packages and activities
- Approved quantities and units
- Weights and weighting basis
- Planned start and finish dates
- Activity calendars and dependencies
- Milestone dates
- Planned progress curve by day/week/month
- Excluded or pending scope
- Linked change, contract or decision record
- Calculation-engine version and timezone

11.2 Baseline Rules

WP-014	Only one Current Approved Baseline may be active for the same project and baseline scope at a time.
WP-015	Activating a new baseline must preserve the Original Baseline and all superseded versions and must regenerate future planned values without changing historical actual progress.
WP-016	Reports must allow comparison against Original Baseline, Current Approved Baseline and Forecast simultaneously.
WP-017	A baseline must fail activation if work-package weights are incomplete, duplicate, negative or outside configured tolerance.
WP-018	Backdated baseline activation must require a reason and must not alter already issued snapshots; revised comparative reports must identify the baseline used.

12. Work Progress Dashboard

12.1 Global Portfolio Dashboard

KPI / widget	Behavior and drill-down
Accepted Physical Progress	Weighted accepted progress across selected projects, with baseline and effective date.
Planned Progress	Planned value from selected baseline and reporting date.
Variance	Actual minus planned in percentage points, plus status classification.
Forecast Completion	Latest forecast finish versus current and original baseline finish.
Delayed Critical Work	Critical or near-critical packages/activities forecast late.
Blocked Work	Packages with unresolved blocking constraints or safety/quality holds.
Stale Progress	Packages not updated within configured cadence.
Unverified Submissions	Submitted quantities or reports awaiting verification.
Contractor Performance	Progress, production trend, delay, rework and reporting timeliness by contractor.
Data Quality	Coverage, freshness, evidence and mapping completeness score.
Progress Claims	Claimed, measured, accepted, certified and disputed values.
Recovery Exposure	Late packages without an active quantified recovery plan.

12.2 Project Dashboard

- S-curve showing original planned, current planned, accepted actual and forecast progress.
- Overall and discipline/location/package progress cards with drill-down.
- Forecast milestone panel with days ahead/behind and confidence.
- Critical path and near-critical activity list.
- Active, delayed, blocked, suspended and substantially complete work packages.
- Current workforce, equipment and production rate summary.
- Material shortage and procurement-delivery risks linked to affected work.
- Open constraints ordered by required-by date and schedule impact.
- Latest daily report and reporting compliance calendar.
- Progress photo timeline and evidence coverage.
- Contractor claimed versus accepted versus certified progress.
- Recent revisions, overrides, corrections and audit events.

12.3 Dashboard Rules

WP-019	Every dashboard KPI must be clickable and produce the exact filtered record set or calculation breakdown behind the number.
WP-020	Dashboard values must display last calculated time, effective progress date, selected baseline and project timezone.
WP-021	Portfolio rollups must exclude inaccessible projects and must not reveal confidential project names or values through counts, charts or tooltips.
WP-022	Cards must use compact premium hierarchy and must not become large decorative blocks that reduce operational density.

13. Work Packages Module Page

13.1 Supported Views

View	Use
Table	Default control view for high-volume scanning, bulk actions and configurable columns.
Board / Kanban	Group by lifecycle, contractor, discipline, building or delay state. Dragging may request a valid transition but cannot bypass validation.
Timeline / Gantt	Show package and activity dates, dependencies, baselines, forecast and milestones.
Location Matrix	Rows by work package/discipline and columns by building/floor/zone, with progress heat cells.
Progress Curve	Compare planned, actual and forecast trends for the current filtered set.
Map / Plan later	Show geospatial or floor-plan progress where location geometry exists.
Saved View	Persist personal or shared filters, grouping, sorting, columns, baseline choice and visualization.

13.2 Default Columns

- Package code and name
- Project and structure/location
- Discipline and category
- Contractor / responsible team
- Lifecycle status
- Readiness status
- Accepted progress
- Planned progress
- Variance
- Baseline start/finish
- Forecast finish
- Days ahead/behind
- Critical/near-critical indicator
- Last accepted update
- Data freshness
- Open blockers
- Evidence coverage
- Claim/certification status
- Personal star

13.3 Search and Bulk Actions

- Create package
- Import packages
- Duplicate as template
- Assign responsible user/team/contractor
- Change configured fields
- Create or update baseline working version
- Create task
- Create constraint or delay
- Request progress update
- Export selected records
- Archive eligible packages
- Save view
- Generate report for selection

WP-023

Bulk progress updates must require a preview that shows calculated impact, validation errors and affected baseline/rollup dimensions before commit.

WP-024

The page must support virtualized rendering and cursor pagination for projects with tens of thousands of packages and activities.

WP-025

Hover-only controls must have keyboard and touch equivalents. Mobile must not require horizontal precision interactions for critical actions.

14. Work Package Page

The Work Package Page is the complete operational workspace for one defined scope of physical work.

14.1 Header

- Package name and code
- Project, structure and location breadcrumb
- Discipline and package type
- Lifecycle and readiness status
- Accepted, planned and forecast progress
- Baseline and forecast dates
- Contractor / responsible owner
- Critical or blocked indicator
- Star / follow action
- Primary action based on state
- More actions menu
- Last update and data-quality indicator

14.2 Sections / Tabs

Section	Content
Overview	Scope, inclusions/exclusions, completion definition, owners, dates, progress, readiness and warnings.
Progress	Measurement method, quantity lines, updates, verification, accepted values, reversals and calculation drill-down.
Schedule	Activities, milestones, dependencies, calendars, baseline/forecast dates, float and lookahead.
Locations	Buildings, floors, zones, units, grid/chainage and progress by location.
Daily Operations	Daily/shift reports, production logs, workforce, equipment, deliveries and issues.
Materials	Required, ordered, delivered, issued, consumed, returned and shortage links from authoritative modules.
Equipment	Assigned assets, hours, output, downtime and availability links.
Contractors	Assigned companies, crews, claims, measured/certified progress and performance.
Constraints and Delays	Open/closed blockers, delay events, notices, mitigation and recovery.
Quality and HSE	Linked inspections, hold points, defects, permits, safety stops and readiness gates.
Documents and Evidence	Drawings, method statements, photos, videos, measurement sheets and linked Document records.
Commercial	BOQ, contract, cost code, budget, variations, claims and certification links.
Tasks and Meetings	Linked coordination records only; does not duplicate task or meeting content.
Activity	Immutable lifecycle, field changes, submissions, acceptances, corrections, links and exports.

14.3 Quick Update Panel

- Effective date/time and shift
- Location
- Measurement lines or milestone
- Completed quantity / percent according to method
- Remaining quantity confirmation
- Crew and equipment references

- Production notes
- Constraints encountered
- Photos/evidence
- Submit as draft or for verification
- Offline save state and sync status

WP-026

The Work Package Page must always show whether displayed progress is submitted, accepted, certified or paid; ambiguous labels such as Progress alone are not permitted.

WP-027

Users must be able to trace a package rollup to each accepted contribution and then to the original field record and evidence.

15. Construction Schedule and Lookahead Planning

15.1 Schedule Entities

- Schedule version
- Calendar and working periods
- Activity
- Milestone
- Dependency
- Constraint date
- Lag / lead
- Baseline dates
- Forecast dates
- Actual dates
- Total and free float
- Critical/near-critical flag
- Progress mapping
- Resource target
- Location and work-package links
- Schedule note and revision reason

15.2 Dependency Types

Type	Definition	Validation
Finish-to-Start (FS)	Successor starts after predecessor finishes.	Default; lag/lead supported.
Start-to-Start (SS)	Successor starts after predecessor starts.	Use for overlapping sequences.
Finish-to-Finish (FF)	Successor finishes after predecessor finishes.	Completion relationship.
Start-to-Finish (SF)	Successor finishes after predecessor starts.	Rare; warning and explicit confirmation required.
External dependency	Dependency to an external milestone, permit, delivery or client action.	Required owner and expected date.
Soft dependency	Planning preference that may be overridden.	Must be visually distinct from hard logic.

15.3 Lookahead Planning

- Configurable 1, 2, 3, 4, 6, 8 or 12-week horizon.
- Activities grouped by location, contractor, discipline or responsible team.
- Readiness checklist for drawings, approvals, permits, access, predecessors, materials, labor, equipment, quality and safety.
- Commitment fields: can start, cannot start, committed finish, reason and owner.
- Weekly plan reliability: planned activities completed, not completed and reason categories.
- Carry-over tracking without deleting missed commitments.
- Constraint removal dates and overdue indicators.
- Lookahead freeze or publication snapshot for historical comparison.
- Calendar and meeting hooks for weekly planning reviews.

15.4 Schedule Calculation Rules

WP-028	The schedule engine must detect dependency cycles before saving and return the exact cycle path.
WP-029	Critical and float calculations must be recomputed after relevant activity, calendar, dependency or forecast changes and stored with calculation timestamp/version.
WP-030	Actual start and finish dates are immutable operational facts. Corrections require an audited correction entry.
WP-031	A package may be physically ahead in percent but still forecast late due to remaining critical sequence; the UI must show both facts.
WP-032	Deleting an activity that has historical progress, baseline or evidence links is prohibited; it may be cancelled or superseded.

16. Progress Measurement Engine

Each work package must use one configured primary measurement method and may use supporting indicators. The method is fixed for an active baseline unless changed through a controlled new baseline or explicitly audited method revision.

Method	Calculation basis	Suitable use	Key controls
Installed Quantity	Accepted installed quantity / current approved scope quantity.	Concrete, steel, masonry, cable, pipe, facade panels, earthworks.	Unit consistency, quantity caps, location uniqueness and reversals.
Completed Units	Accepted completed units / total units.	Apartments, rooms, piles, doors, fixtures or repetitive assets.	Unique unit IDs and completion definition.
Weighted Milestones	Sum of completed/partially completed milestone weights.	Design, procurement, commissioning and complex packages.	Milestone weights total 100 percent and evidence per milestone.
Activity Weighted	Weighted accepted progress across linked schedule activities.	Packages controlled through detailed schedule.	No duplicate activity allocation; baseline mapping required.
Duration Based	Elapsed approved productive duration / planned duration.	Only where duration reasonably represents value.	Cannot advance during suspension or nonproductive time unless configured.
Cost Weighted	Earned or accepted cost value / package budget.	High-level portfolio comparison where physical measure is unavailable.	Finance-linked values; never presented as pure physical progress.
Manual Physical Percent	Expert-entered percent with reason and evidence.	Exceptional or non-quantifiable scope.	Configured authority, mandatory rationale, change limit and enhanced audit.
Composite Method	Controlled combination of sub-methods with weights.	MEP systems, commissioning, design-build packages.	Sub-method weights total 100 percent; transparent rollup.

16.1 Progress Update Record

- Permanent update ID and sequence
- Work package and baseline
- Effective date/time, project timezone and shift
- Submitter, company, role and device context
- Measurement method and lines
- Location and scope dimension
- Previous accepted cumulative value
- Current-period quantity/value
- New cumulative submitted value
- Evidence and document references
- Crew, material and equipment references
- Constraint/delay flags
- Verification status and verifier
- Acceptance/rejection reason
- Correction or reversal linkage
- Audit and offline synchronization metadata

16.2 Verification States

Draft	Submitted	Under Verification	Accepted	Rejected
Returned for Correction is a non-terminal review state. Accepted updates are immutable and corrections create linked reversal/correction records.				
WP-033	Only Accepted progress updates contribute to official physical progress. Submitted values may be displayed separately as pending.			
WP-034	Accepted cumulative quantity must not exceed the current approved scope quantity except through a configured overrun state linked to pending or approved change scope.			
WP-035	A progress update must specify effective date separately from creation timestamp. Backdated effective dates require a reason and policy check.			
WP-036	The same location, unit, element or quantity segment cannot be accepted more than once for the same scope dimension.			
WP-037	Manual progress must be clearly labeled, must carry mandatory evidence/rationale and may be limited by configured per-update change thresholds.			

17. Progress Rollups, Weighting and Formula Registry

17.1 Canonical Formula Concepts

$\text{Quantity Progress \%} = \text{Accepted Cumulative Quantity} / \text{Current Approved Scope Quantity} \times 100$
 $\text{Weighted Actual Progress \%} = \text{Sum}(\text{Child Weight} \times \text{Child Accepted Progress}) / \text{Sum}(\text{Included Child Weights})$
 $\text{Weighted Planned Progress \%} = \text{Sum}(\text{Child Weight} \times \text{Child Planned Progress at Report Date}) / \text{Sum}(\text{Included Child Weights})$
 $\text{Progress Variance (percentage points)} = \text{Accepted Actual Progress} - \text{Planned Progress}$
 $\text{Forecast Delay (days)} = \text{Forecast Finish Date} - \text{Current Baseline Finish Date}$
 $\text{Production Rate} = \text{Accepted Quantity} / \text{Productive Time or Reporting Period}$
 $\text{Remaining Duration} = \text{Remaining Quantity} / \text{Forecast Production Rate, adjusted for calendar and sequence}$
 $\text{Percent Plan Complete (PPC)} = \text{Completed Planned Commitments} / \text{Total Planned Commitments} \times 100$

17.2 Weighting Bases

Weighting basis	Behavior	Warning
Quantity	Weight derived from standardized quantity.	Only comparable units or normalized factors may roll up directly.
Budget / Contract Value	Weight derived from approved financial value.	Label as cost-weighted; may not represent physical effort.
Labor Hours	Weight derived from planned labor.	Requires reliable labor plan and updates.
Duration	Weight derived from baseline duration.	Can distort value for waiting/low-effort activities.
Custom Engineering Weight	Explicit approved weight by package/activity.	Must have rationale and total validation.
Composite	Configured combination of quantity, value, labor or milestones.	Formula and version must be visible and reproducible.

17.3 Rollup Controls

- Weights must be versioned with the baseline.
- Excluded, cancelled or pending-change scope must have explicit rollup treatment.
- Negative weights are prohibited.
- Rollup weight totals must equal the configured expected total within tolerance before baseline activation.
- Parent progress must be calculated from children unless the parent uses a separate approved measurement method.
- Rounding occurs only for presentation; calculations use configured precision.

- No child may contribute through two independent paths to the same parent rollup.
- Each snapshot stores formula version, input IDs, baseline ID and calculation timestamp.
- Historical reports must use the formula and scope effective at that time.

17.4 Optional Earned Value Indicators

Planned Value (PV) = Budget at Completion x Planned Progress
 Earned Value (EV) = Budget at Completion x Accepted Physical Progress
 Schedule Variance (SV) = EV - PV
 Schedule Performance Index (SPI) = EV / PV
 Cost Variance (CV) = EV - Actual Cost
 Cost Performance Index (CPI) = EV / Actual Cost

Important boundary

Earned-value indicators are optional analytical views. Finance remains authoritative for actual cost and budget. Contract certification remains separate from earned physical value. Divide-by-zero, missing-cost and incompatible-scope cases must return Not Available, not misleading zeros.

WP-038

Every rollup result must be reproducible from stored inputs and the calculation-engine version.

WP-039

The system must reject a configuration that produces circular rollups or allows a child contribution to reach the same parent more than once.

WP-040

Reports must use percentage points for actual-versus-planned difference and must not label that value as percent difference unless explicitly calculated.

18. Daily Site Reports and Site Diary

The Daily Site Report is the controlled operational record of one project/site day or shift. It must be fast enough for field use and detailed enough for later dispute, productivity and delay analysis.

18.1 Report Scope and Identity

- Project and site
- Report date and project timezone
- Shift and working calendar
- Report number and revision
- Prepared by, contributors and responsible manager
- Weather source and manual observations
- Working/non-working day state
- Submission deadline and status
- Related work packages, locations and contractors

18.2 Report Sections

Section	Required capabilities
Weather and site conditions	Temperature, precipitation, wind, visibility, extreme conditions, ground condition, working impact and source.
Workforce	Company/contractor, crew, trade, headcount, planned hours, actual hours, productive hours, overtime and absence reason.
Equipment	Asset, operator, working/idle/downtime hours, meter readings, fuel link, output and downtime cause.
Work completed	Package, location, activity, current-period quantity, cumulative quantity, narrative and evidence.
Deliveries and logistics	Supplier, PO/delivery link, material, expected/received quantity, status, rejection and affected work.
Materials used	Inventory issue/consumption links, quantity, location and waste/rework reason.

Section	Required capabilities
Issues and constraints	Description, category, owner, required-by date, schedule impact and task/delay link.
Quality	Inspection links, hold points, tests, defects, rejected work and rework.
HSE	Permit links, toolbox talks, incident/near-miss links, stoppages and observations.
Visitors and instructions	Visitor, organization, purpose, arrival/departure and formal instruction/document link.
Photos and videos	Time, uploader, location, orientation, work package, annotations and caption.
Tomorrow / next shift plan	Planned activities, required resources, unresolved prerequisites and commitments.
Sign-off	Prepared/submitted timestamps, verifier, comments, locked revision and correction history.

18.3 Daily Report Lifecycle

Draft	In Preparation	Submitted	Verified	Locked
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Returned for Correction and Superseded by Revision are supported states. A locked report is never edited in place.

18.4 Daily Report Rules

WP-041	One project may have multiple reports per date only when the shift, site area or reporting scope is explicitly different.
WP-042	A submitted or locked report correction must create a numbered revision that preserves the original report, signatures and calculation impact.
WP-043	Weather, labor, equipment and progress entries must preserve both source timestamp and user-entered effective period.
WP-044	The system must identify missing daily reports, late submissions, incomplete mandatory sections and inconsistent totals.
WP-045	Daily report PDF generation must produce a stable, versioned document linked to the report and Documents Module without duplicating the underlying record.

19. Field Capture, Mobile and Offline Operation

19.1 Critical Mobile Actions

- Open recent project/work package
- Start or continue daily report
- Capture completed quantity
- Scan QR/barcode to identify location, asset or material
- Capture photo/video and annotate
- Log crew and equipment
- Create blocker/constraint
- Record delivery exception
- Save draft offline
- Review sync queue and conflicts
- Submit for verification when online

19.2 Offline Data Model

- Client-generated UUIDs for offline-created records.
- Local encrypted project cache limited to authorized projects and retention window.

- Append-only local mutation queue with dependency order.
- Idempotency key per mutation.
- Attachment upload state with checksums, chunking and resumable upload.
- Local reference snapshots for packages, locations, units, crews, assets and products.
- Explicit Pending Sync, Synced, Conflict and Failed states.
- Automatic retry with exponential backoff and manual retry/cancel controls.
- No background submission of a record the user left as Draft.

19.3 Field Evidence Metadata

Metadata	Rule
Capture timestamp	Device time plus server receipt time; clock difference recorded.
Location	Project/LBS reference required where configured; GPS optional and privacy-controlled.
Device/user	User, session and device identifier stored for audit.
File integrity	Checksum and original metadata retained; derived renditions are separate.
Orientation / annotation	Optional compass, floor-plan pin, drawing markup or model element link.
Offline state	Capture and local save timestamps retained through synchronization.
Privacy	Location and device data collected only where configured and lawful; no covert tracking.

WP-046

Offline submission must never create duplicate progress, report, photo or quantity records after retry. APIs must enforce idempotency.

WP-047

The UI must show the user which records and attachments are not yet synchronized and must not imply server acceptance before confirmation.

WP-048

GPS must remain optional, visible to the user and configurable by company/project policy.

20. Workforce, Crews and Productivity

20.1 Crew Architecture

- Crew ID, name and trade
- Employer/contractor
- Foreman/supervisor
- Members or headcount-only mode
- Qualifications and certifications references
- Assigned project, package, location and shift
- Planned and actual hours
- Productive, idle, waiting, rework and overtime hours
- Output quantity and unit
- Labor notes and evidence

20.2 Productivity Metrics

Labor Productivity = Accepted Output Quantity / Productive Labor Hours
 Labor Utilization = Productive Labor Hours / Paid or Logged Labor Hours
 Rework Ratio = Rework Labor Hours / Total Labor Hours
 Crew Plan Attainment = Actual Accepted Output / Planned Output

Metric rule	Requirement
Source separation	HR/Attendance owns employment and attendance; Work Progress owns package/shift allocation and productive context.
Headcount mode	Allow aggregate crew logs when individual worker data is unavailable or restricted.

Metric rule	Requirement
Individual mode	Support worker-level logs only when permitted and necessary.
Privacy	Salary, health and disciplinary data must never appear in Work Progress.
Qualifications	Read-only qualification checks may warn or block assignment when future HR/HSE integration requires it.
Historical role	Retain contractor, trade and role snapshot at the time of work.

WP-049	Workforce productivity must not be used as an automated employee performance judgment. It is an operational package/crew indicator requiring context.
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WP-050	Labor totals in a daily report must reconcile with its crew/shift lines within configured tolerance or display a validation warning.
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21. Materials, Deliveries and Consumption

Work Progress must explain whether material availability and consumption support the reported physical progress while preserving Inventory and Procurement as authoritative sources.

Material state	Authoritative source	Work Progress behavior
Required quantity	Work Progress / package planning	Stores planned requirement by package, location and period.
Requested / ordered	Procurement	Displays linked request, PO, supplier and expected delivery.
Received / rejected	Inventory receipt / Procurement context	Displays authoritative receipt status and delivery exception.
Available stock	Inventory	Read-only availability by warehouse/site.
Issued to project/package	Inventory	Links stock issue/movement to package and daily report.
Consumed / installed context	Inventory movement plus Work Progress allocation	Stores operational allocation and accepted installed quantity relationship.
Returned / transferred	Inventory	Displays authoritative movement.
Waste / damage / rework	Inventory/QA/QC/Work Progress links	Records reason and affected package without altering stock ledger directly.
Material on site for claim	Finance/Contracts/Inventory	Provides evidence and quantity references; certification logic remains controlled.

21.1 Material Readiness

- Required-by date
- Planned quantity
- Available quantity
- Ordered quantity
- Expected delivery
- Quality approval state
- Storage/location readiness
- Shortage quantity
- Affected activities/packages
- Responsible buyer/logistics owner
- Mitigation and alternative material link

WP-051	Work Progress must never write stock balances directly. All stock-changing actions must call Inventory APIs and store the returned authoritative movement ID.
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WP-052	Installed quantity may not automatically equal issued quantity. Waste, storage, unused material and timing differences must remain visible.
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WP-053 A material shortage must be linkable to one or more constraints, delays and forecast impacts.

22. Equipment and Asset Utilization

Operational record	Fields / behavior
Assignment	Asset, project, package, location, operator, start/end and assignment source.
Shift usage	Working, idle, setup, standby and downtime hours.
Meter reading	Start/end reading and authoritative Asset record link.
Production	Output quantity and unit linked to work package.
Downtime	Reason, start/end, responsible provider, work order and affected activity.
Availability constraint	Needed-by date, availability status and alternative asset.
Internal/external cost	Read-only Finance/Assets rate or cost allocation link.
Inspection/qualification	Read-only inspection and operator qualification status where required.

Equipment Utilization = Working Hours / Available Assigned Hours
 Equipment Productivity = Accepted Output Quantity / Working Equipment Hours
 Downtime Ratio = Downtime Hours / Available Assigned Hours

WP-054 Asset identity, meter history, maintenance and ownership remain authoritative in Assets. Work Progress stores allocation context and source links only.

WP-055 A package progress update may reference equipment output, but equipment hours alone cannot increase physical progress.

23. Subcontractor Progress and Commercial Measurement

23.1 Separation of Progress States

State	Meaning	Contributes to
Reported by Contractor	Contractor-submitted quantity/percent.	Pending view only.
Measured by Site Team	Quantity physically measured or jointly recorded.	Measurement comparison; not automatically official.
Accepted Physical Progress	Verified quantity accepted for operational progress.	Official physical progress rollup.
Claimed Commercial Progress	Value/quantity included in contractor application.	Commercial claim view.
Certified Progress	Quantity/value certified under contract process.	Certification and Finance readiness.
Paid Progress	Finance payment state.	Payment reporting only.

23.2 Contractor Package Workspace

- Assigned scope and locations
- Contract and BOQ links
- Baseline and lookahead
- Daily reports and attendance submissions
- Submitted progress and evidence
- Measured and accepted quantities
- Claims and certification history
- Open rejections and disputes
- Delays attributable / non-attributable context

- Quality and HSE links
- Materials and equipment provided by each party
- Performance indicators and reporting compliance

23.3 Contractor Performance Indicators

- Planned versus accepted progress
- Forecast delay
- Production-rate trend
- Lookahead commitment reliability
- Reporting timeliness
- Rejected measurement rate
- Rework and defect links
- Constraint response time
- Safety/quality hold impact
- Claimed-to-accepted variance
- Accepted-to-certified variance

WP-056

The system must never overwrite contractor-submitted quantities with measured or accepted quantities. Each state remains independently auditable.

WP-057

Contractor performance must always disclose reporting period, project/package scope and source completeness.

24. Delays, Constraints, Risks and Recovery Planning

24.1 Constraint Register

Field	Requirement
Identity	Constraint ID, title, category, status and priority.
Scope	Project, package, activity, location, milestone and affected parties.
Cause	Drawing, approval, permit, material, equipment, labor, access, predecessor, design, client, weather, quality, HSE, finance, external or custom.
Ownership	One accountable owner, responsible team/company and escalation contact.
Dates	Identified date, required-by date, forecast release, actual release and aging.
Impact	Blocking/non-blocking, affected quantity, potential delay, cost/contract links and critical-path relevance.
Resolution	Action, task links, documents, evidence, decision/approval hook and release verification.
History	Status changes, due-date changes, comments, reasons and immutable activity.

24.2 Delay Event

- Delay ID and type
- Start/end or ongoing
- Affected package/activity/milestone/location
- Observed facts
- Cause and contributing causes
- Responsible/attribution party as allegation or confirmed classification
- Calendar days and working days
- Critical path / float impact
- Concurrent delay links
- Notice/contract record links
- Mitigation and recovery actions
- Evidence
- Status and closure reason

24.3 Delay Lifecycle

Open	Under Investigation	Impact Assessed	Mitigation Planned	Under Recovery	Resolved	Closed
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Cancelled and Merged are controlled states. Closure does not delete historical schedule impact.

24.4 Recovery Plan

Area	Fields
Target	Affected milestones/packages and target recovery date.
Actions	Resequencing, added crew, overtime, second shift, additional equipment, alternative material, design change or scope split.
Resources	Incremental labor, equipment, material and subcontractor requirements.
Schedule	Scenario dates, dependency changes, remaining duration and recovered days.
Commercial	Estimated incremental cost and contract/change links.
Monitoring	Daily/weekly target output, owner, review cadence and variance.
Outcome	Actual recovered days, cost, lessons learned and closure.

WP-058 Closing a constraint requires release evidence or an explicit waiver reason and actor.

WP-059 A delay duration must not be calculated solely from issue open age. Schedule impact, calendars, concurrency and actual affected dates are separate fields.

WP-060 Recovery scenarios must not alter the official forecast or baseline until explicitly promoted through the configured control point.

WP-061 The UI must distinguish factual observations, alleged responsibility and formally determined responsibility.

25. Evidence, Progress Photos and Location Capture

25.1 Evidence Types

- Photo
- Video
- 360-degree image
- Drone capture reference
- Marked-up drawing
- Measurement sheet
- Survey record
- Test/inspection reference
- Delivery ticket
- Timesheet/crew log
- Equipment log
- Model element screenshot/reference
- Signed site instruction
- Other controlled Document record

25.2 Progress Photo Record

- Original file and Document/media reference
- Thumbnail and optimized rendition
- Capture and upload timestamps
- Project timezone
- Uploader and organization
- Project/package/activity/location links
- Building/floor/zone/unit/grid/chainage
- Optional GPS and compass

- Caption and tags
- Before/after relationship
- Annotation shapes and text
- Confidentiality and visibility classification
- Duplicate/perceptual-hash warning
- Evidence acceptance state
- Retention and legal hold flags

25.3 Evidence Coverage

The module may calculate a rule-based evidence coverage indicator based on whether configured packages, milestones, quantities, locations or reporting periods have valid linked evidence. This is a data-completeness indicator, not proof that work is correct.

WP-062	Deleting a media rendition must not delete the original controlled file. Deleting an original evidence record after acceptance requires retention and audit policy checks.
WP-063	Photo timestamps, GPS and metadata must be displayed as captured metadata, not treated as infallible proof. Manual corrections must remain separately visible.
WP-064	Progress photos must support chronological, location, package, contractor and before/after views.

26. Quality, HSE and Readiness Gates

QA/QC and HSE are separate future modules. Work Progress must consume their states as gates and context without reproducing their full workflows.

Gate type	Examples	Effect
Information readiness	Approved-for-construction drawing, method statement, RFI response, approved sample.	May block Ready or progress acceptance.
Quality hold point	Inspection request, test result, witness point, NCR closure.	May block continuation, acceptance or completion.
Safety readiness	Permit to work, risk assessment, induction, lifting plan, safety clearance.	May block start or continue state.
Access readiness	Area released, predecessor completed, temporary works/access available.	May block Ready.
Material readiness	Approved material, quantity available, storage and test certificates.	May block Ready or planned output.
Equipment readiness	Required asset available, inspected, calibrated and qualified operator available.	May block Ready.
Client/consultant decision	Approved change, sample, mock-up or instruction.	May block affected scope.
Contractual prerequisite	Notice, access date, milestone condition or authority permit.	May affect readiness and delay context.

26.1 Gate Configuration

- Required or advisory
- Applicable package type/discipline
- Source module and record type
- Required state
- Required-by stage
- Blocking behavior
- Override capability key
- Override expiry
- Evidence/document requirement
- Recheck event and stale-state handling

WP-065	A blocked quality or safety gate must be visible on package, schedule, lookahead and dashboard views.
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WP-066	A Work Progress user cannot close or alter authoritative QA/QC or HSE records from this module.
WP-067	Gate overrides must identify actor, reason, scope and expiry and must be reviewable in audit reports.

27. Progress Claims, Measurement Sheets and Certification

27.1 Measurement Sheet

Field group	Fields
Identity	Sheet number, period, project, contract, contractor, package and status.
Basis	BOQ/contract lines, drawing revision, measurement method and currency where relevant.
Lines	Location, description, unit, contract quantity, previous quantity, current claimed, measured, accepted, certified and cumulative quantities.
Value	Rate, gross value, variations, materials on site, deductions, retention context and tax reference where applicable.
Evidence	Photos, survey, marked drawings, inspection links, delivery/stock references and attachments.
Review	Submitter, measurer, reviewer, comments, disputed quantity and reasons.
Outcome	Certified quantity/value, rejected quantity, carry-forward and Finance/Contract links.
Audit	Revision, signatures/decision hooks, timestamps and immutable activity.

27.2 Measurement Sheet Lifecycle

Draft	Submitted	Measured	Under Review	Agreed / Disputed	Certified	Closed
Rejected, Returned and Superseded by Revision are supported states. Certification decision routing is delegated to the future Approvals Engine.						

27.3 Certification Controls

- Previous certified cumulative values are immutable.
- Current-period certified quantity cannot exceed accepted/measured scope unless an authorized exception is linked.
- Duplicate contract/BOQ/location/period claims are detected.
- Certified physical quantity and certified monetary value remain related but distinct.
- Retention, advance recovery, taxes and payment are Finance/Contracts concerns displayed through links.
- Disputed quantities remain visible and may carry forward without being silently discarded.
- Certification PDF/document snapshots are versioned and linked to Documents.
- Corrections use a revised certificate, credit/debit adjustment or explicit reversal according to connected-module rules.

WP-068	Contractor claim, site measurement, accepted physical progress and certified commercial progress must remain separate fields and histories.
WP-069	A certification event must publish a versioned payload to Finance/Contracts without writing directly to their ledgers or legal terms.
WP-070	The system must show the reason for every difference between claimed, measured, accepted and certified quantity.

28. Cost, Finance, Contracts and Change Integration

28.1 Finance Integration

- Budget and cost-code context
- Committed cost
- Actual cost by project/package/period
- Labor/equipment/material cost summaries where authorized
- Earned-value analytical inputs
- Progress certification readiness
- Forecast cost impact from recovery plans
- Read-only invoice, payment and cash-flow state
- Correlation from cost transaction to package/activity where available

28.2 Contracts Integration

- Contract/subcontract parties
- Scope and BOQ links
- Contract start/finish and milestones
- Payment schedule and measurement rules
- Notice and instruction links
- Variation/change records
- Delay/claim evidence
- Retention and guarantee context
- Contractor progress claim links

28.3 Change and Variation Treatment

Scope state	Progress treatment
Original approved scope	Included in original baseline and rollups.
Approved variation	Added through a new scope/baseline version or controlled package change.
Pending variation	Tracked separately; may record performed-at-risk quantity but cannot silently alter approved denominator.
Omitted scope	Removed through controlled change; historical completed quantity remains visible and rollup treatment is versioned.
Rework	Recorded separately from original accepted completion; does not automatically add net progress.
Overrun quantity	Flagged beyond approved scope and linked to pending/approved change decision.
Client change / redesign	May suspend or split affected package and preserve prior work/evidence.

WP-071

Pending variation work must never silently increase the denominator or official baseline progress. It must be visible as separate at-risk scope.

WP-072

Actual cost values displayed in Work Progress must include source currency, conversion basis where applicable, effective date and Finance record link.

WP-073

A contract milestone may be marked operationally achieved only with linked evidence and must remain separate from formal contractual acceptance.

29. Forecasting, Completion Confidence and Data Quality

29.1 Forecast Inputs

- Remaining accepted scope quantity
- Recent production rates
- Crew and equipment plan

- Working calendars
- Dependencies and float
- Open constraints and forecast release dates
- Material/procurement availability
- Quality/HSE gates
- Weather or seasonal assumptions where configured
- Recovery plan actions
- Pending changes and excluded scope
- Manual planner forecast with rationale

29.2 Forecast Methods

Method	Behavior
Schedule logic forecast	Recalculate dates from remaining durations, calendars and dependencies.
Quantity-rate forecast	Remaining quantity divided by selected production rate, then sequenced through calendar/dependencies.
Planner forecast	Authorized date with reason, assumptions and comparison to calculated forecast.
Scenario forecast	What-if resources, shifts, sequence or scope; no effect on official forecast until promoted.
Hybrid forecast	Use calculated date with documented planner overrides for selected activities.

29.3 Completion Confidence

A rule-based completion confidence indicator may summarize data quality and forecast stability. It must not pretend to be an AI prediction. The score must disclose its factors and never replace the forecast date.

- Update freshness
- Accepted-measurement coverage
- Evidence coverage
- Schedule mapping completeness
- Production-rate sample size
- Constraint due-date quality
- Critical-path stability
- Pending variation exposure
- Manual override frequency
- Unresolved data conflicts

29.4 Data Quality Score

Dimension	Example checks
Completeness	Missing package scope, owner, baseline, method, location, quantity or contractor.
Freshness	Days since accepted update or daily report.
Consistency	Daily quantity versus progress update, quantity versus scope, dates versus lifecycle.
Evidence	Required evidence missing or unlinked.
Mapping	Unmapped BOQ, activity, location, cost or contract references.
Verification	High pending/rejected ratio or unresolved corrections.
Baseline health	Missing/invalid weights, stale working schedule or uncontrolled date changes.
Offline health	Long-unsynced records or repeated conflicts.

WP-074 Forecast calculations must store assumptions, input period, production-rate source and calculation version.

WP-075 A confidence or data-quality score must be decomposable into visible factors and cannot be used to hide missing information.

WP-076

When insufficient data exists, the system must return Insufficient Data rather than fabricate a completion forecast.

30. Search, Filters, Saved Views and Reporting

30.1 Search

- Package code/name/description
- Project, structure, floor, zone, unit or chainage
- Contractor, company, team or responsible user
- Activity/milestone
- BOQ/contract line
- Constraint/delay ID and text
- Daily report number/date
- Measurement/certificate number
- Document filename/metadata through Documents search
- Photo caption/tag
- External reference or import ID

30.2 Filters

- Company/group
- Project
- Project status
- Structure/location
- Discipline/category
- Package lifecycle/readiness
- Contractor
- Responsible user/team
- Measurement method
- Progress range
- Variance range
- Critical/near-critical
- Blocked/delayed/stale
- Date range
- Baseline
- Evidence coverage
- Data quality
- Claim/certification status
- Saved personal/shared views

30.3 Standard Reports

Report	Minimum content
Daily Site Report	Complete day/shift record with revisions and evidence.
Weekly Progress	Planned/actual/forecast, quantities, critical work, constraints, labor/equipment and next-week plan.
Monthly Progress	Executive summary, S-curve, package/location/discipline status, contractor performance, commercial measurement and risks.
Client Progress	Authorized progress, milestones, photos, delays and decisions without confidential internal fields.
Work Package Report	Scope, schedule, progress history, resources, constraints, evidence and commercial links.
Delay and Constraint Report	Aging, impact, owner, release forecast, schedule links and recovery status.
Productivity Report	Output versus labor/equipment by package, crew, contractor and period.
Subcontractor Report	Planned, reported, measured, accepted, certified and disputed values.
Lookahead / PPC Report	Commitments, readiness, completion and reason analysis.
Progress Data Quality Report	Missing, stale, inconsistent, unverified and conflicting records.

Report	Minimum content
Baseline Comparison	Original, current baseline, actual and forecast dates/progress.
Audit Report	Selected record changes, corrections, overrides, exports and actor history.

30.4 Export Rules

- PDF and Excel export
- CSV for structured data where permitted
- Report effective date and baseline identity
- Applied filters and timezone
- Generated by and generated at
- Data classification / confidentiality
- Stable report ID and checksum
- Document link and revision where officially issued
- Asynchronous generation for large reports

WP-077	Reports must disclose whether values are accepted, submitted, claimed, certified or paid.
WP-078	Saved views must preserve layout, filters, sort, grouping, visible columns, baseline selection and reporting period per user.
WP-079	Large exports must run asynchronously and must not block interactive application requests.

31. Notifications, Calendar and Collaboration Hooks

The central Notifications, Calendar, Meetings and Approvals modules will be designed later. This module must publish structured events and consume their returned identifiers without implementing their full engines.

Trigger	Suggested event / hook
Daily report due or late	Notification/reminder event with project, date, shift and responsible user.
Progress update pending verification	Verification request payload.
Package forecast late	Risk notification with milestone and days late.
Constraint approaching required-by date	Reminder/escalation event.
Critical constraint overdue	High-priority event and optional task creation.
Baseline activation/rebaseline	Future approval request payload and calendar milestone update.
Lookahead publication	Calendar/review meeting hook.
Work package start/finish forecast change	Calendar event update.
Measurement/certification submitted	Future approval/review event.
Stale progress or missing evidence	Data-quality notification.
Offline sync failure	User-facing operational notification.
Recovery review due	Calendar/meeting/task hook.

WP-080	Notification event payloads must be idempotent and include tenant, company, project, record, event type, severity, actor and correlation ID.
WP-081	Disabling a future integration through Platform Configuration must remove dead links and fall back to an internal status or activity record where necessary.

32. Documents and Record Folder Architecture

Every project and work package must use the Documents Module through references and shortcuts. Files are not copied into a separate Work Progress storage tree.

32.1 Recommended Project Folder Structure

- 01 Work Progress / 01 Baselines and Schedules
- 01 Work Progress / 02 Work Packages
- 01 Work Progress / 03 Daily Site Reports
- 01 Work Progress / 04 Progress Photos and Videos
- 01 Work Progress / 05 Measurement Sheets
- 01 Work Progress / 06 Contractor Claims
- 01 Work Progress / 07 Progress Certificates
- 01 Work Progress / 08 Delays and Constraints
- 01 Work Progress / 09 Recovery Plans
- 01 Work Progress / 10 Weekly and Monthly Reports
- 01 Work Progress / 11 Closeout
- 01 Work Progress / Transit

32.2 Document Linking Rules

- One Document record may appear by shortcut/reference in project, package, contractor, contract and report contexts.
- Latest approved drawing state is read from Documents; Work Progress stores the revision ID used for measurement/evidence.
- Generated report PDFs become versioned Document records or generated artifacts linked to the source record.
- Photo/video originals may use the Documents/media infrastructure with optimized renditions.
- Document deletion, archive, retention, legal hold and revision remain governed by Documents.
- Broken references must be detectable and repairable without losing the Work Progress record.

WP-082

A progress measurement that depends on a drawing or method statement must retain the exact document revision ID, not only the filename.

WP-083

The same file must not be uploaded separately to Work Progress and Documents. The system must create one authoritative file record and multiple references.

33. Responsive UX, Premium Motion and Accessibility

33.1 Desktop

- Fixed platform shell with project identity and breadcrumb
- Compact KPI strip and exception-first dashboard
- Dense virtualized tables with column presets
- Resizable timeline/Gantt panels
- Side drawers for quick update and drill-down
- Sticky package header and contextual action bar
- Comparison panels for baseline/actual/forecast
- Split views for measurement lines and evidence

33.2 Tablet

- Adaptive two-column package workspace
- Touch-friendly timeline controls
- Bottom-sheet filters and quick actions
- Camera and file capture integration
- Offline queue visible from shell
- No information hidden solely behind hover

33.3 Mobile Web

- Project/package quick switch
- Sticky status/progress header
- One-handed quantity, crew, equipment, photo and constraint capture
- Step-based daily report sections

- Offline-first drafts
- Large touch targets and minimal mandatory typing
- Barcode/QR scanning where browser capability exists
- Conflict resolution cards
- Compact read-only dashboard summaries

33.4 Premium Motion

- Subtle card elevation and restrained image zoom on hover/focus.
- Animated progress changes only after server confirmation; never animate misleading optimistic official values.
- Smooth timeline expansion, filter transitions and drill-down panels.
- Skeleton loading matching final layout.
- Number transitions for KPI updates with reduced-motion alternative.
- Clear sync-state animations without continuous distracting motion.
- No decorative animation in field capture that slows repeated entry.
- All motion respects prefers-reduced-motion.

33.5 Accessibility

- WCAG-aligned contrast and keyboard navigation
- Accessible names for charts, icons and progress states
- Data tables with headers and row context
- Text alternatives for color-coded schedule/progress states
- Screen-reader announcements for sync, validation and status changes
- Focus restoration after drawers/modals
- Touch target minimums
- Chart data downloadable in tabular form
- No critical action dependent on drag-and-drop

WP-084

Red/amber/green status must always include text/icon meaning and must not rely on color alone.

WP-085

Mobile daily reporting must remain usable at 320 px width and under intermittent connectivity.

34. Platform Configuration and Localization

34.1 Configurable Elements

- Module/page/section/widget/action enablement
- Work package types and code formats
- Disciplines and categories
- Lifecycle labels and allowed transitions
- Readiness gates
- Measurement methods and precision
- Weighting bases and tolerances
- Baseline policies
- Daily report templates and mandatory fields
- Shift types and work calendars
- Weather fields and sources
- Constraint/delay categories
- Reason codes
- Evidence requirements
- Progress acceptance thresholds
- Data freshness thresholds
- Confidence/data-quality factors
- Report templates
- Notification event hooks
- Import mappings
- Custom fields and custom dimensions
- Units of measure and conversions

34.2 Localization

- Multiple languages
- Project/company timezone
- Locale-specific date and number formatting

- Metric and imperial units
- Unit conversion with preserved original value
- Multiple currencies for linked commercial views
- Working-week and holiday calendars
- Right-to-left layout readiness where platform supports it

WP-086

Configuration changes that affect calculations must be versioned and applied prospectively unless a controlled recalculation is explicitly requested and audited.

WP-087

Disabling a section or feature must remove navigation, actions, empty placeholders, background jobs and API exposure according to platform policy.

35. Functional Data Model

35.1 Core Entities

Entity	Purpose
WorkProgressProjectConfig	Project-level module settings, baseline policy, cadence and calculation configuration.
WorkPackage	Primary physical scope record.
WorkPackageLocation	Many-to-many package/location allocation.
WorkPackageBOQLink	BOQ/quantity allocation and source reference.
WorkPackageContractLink	Contract/subcontract and scope link.
WorkPackageCostLink	Finance cost center/code/budget link.
WorkPackageDocumentLink	Document/revision relationship.
WorkPackageModelLink	Future BIM/model element relationship.
ScheduleVersion	Working, baseline, rebaseline, forecast or scenario schedule.
ScheduleCalendar	Working periods, holidays and shifts.
ScheduleActivity	Operational activity or milestone.
ActivityDependency	Predecessor/successor relation and lag.
BaselineSnapshot	Immutable baseline activation payload.
ProgressMethodConfig	Configured measurement method and rules.
ProgressUpdate	Field progress submission header.
ProgressUpdateLine	Quantity, unit, milestone or location contribution.
ProgressVerification	Verification/acceptance decision record.
ProgressCorrection	Reversal or correction link.
ProgressSnapshot	Immutable calculation result at a reporting point.
DailySiteReport	Daily/shift report header and lifecycle.
DailyReportRevision	Immutable revision chain.
DailyWeatherLog	Weather and site condition record.
DailyLaborLog	Crew/headcount/hour record.
DailyEquipmentLog	Asset usage/downtime record.
DailyDeliveryLog	Procurement/inventory delivery context.
DailyMaterialLog	Material issue/consumption context.
DailyWorkLog	Completed work narrative and quantity links.
DailyIssueLog	Issue, instruction, visitor or next-plan record.
Crew	Operational crew definition.
CrewAssignment	Crew-to-package/location/shift assignment.
Constraint	Readiness blocker or prerequisite.

Entity	Purpose
ConstraintStatusHistory	Immutable lifecycle events.
DelayEvent	Observed delay and impact record.
RecoveryPlan	Quantified recovery scenario/plan.
RecoveryAction	Action, resource, target and result.
ProgressEvidence	Evidence metadata and source link.
PhotoAnnotation	Markup and location annotation.
SubcontractorProgressSubmission	Contractor-reported progress.
MeasurementSheet	Commercial/quantity measurement header.
MeasurementLine	Claimed, measured, accepted and certified quantities.
ProgressCertificate	Versioned certification result and links.
LookaheadPlan	Published planning horizon snapshot.
LookaheadCommitment	Activity commitment and result.
WorkProgressSavedView	Personal/shared view configuration.
ImportJob	Import mapping, validation and result.
OfflineMutation	Client mutation queue and sync result.
WorkProgressActivity	Immutable audit-oriented event stream.

35.2 Common Technical Fields

- id UUID/ULID
- tenant_id
- owning_company_id
- project_id
- created_at / created_by
- updated_at / updated_by
- record_version
- status
- effective_at
- source_module / source_record_id
- correlation_id
- external_id / import_batch_id
- archived_at / archived_by
- metadata_json for controlled extension only

35.3 Data Integrity Principles

- Tenant and company scope on every record
- Project scope on every execution record
- Foreign keys or validated external references
- Append-only accepted measurements and issued snapshots
- Soft archive instead of destructive delete for historical records
- Decimal numeric types with unit/precision metadata
- Project timezone stored with effective dates
- Idempotency and unique constraints for imports/offline sync
- Outbox pattern for cross-module events
- Audit payloads avoid storing secrets or unrestricted document content

36. Lifecycle and Status Models

Record	Primary states
Work Package	Draft, Planned, Ready, In Progress, Suspended, Substantially Complete, Complete, Closed, Cancelled, Archived
Progress Update	Draft, Submitted, Under Verification, Returned, Accepted, Rejected, Reversed

Record	Primary states
Daily Site Report	Draft, In Preparation, Submitted, Returned, Verified, Locked, Superseded
Schedule Version	Draft, Calculating, Valid, Active Baseline, Superseded, Scenario, Archived
Activity	Not Started, Ready, In Progress, Blocked, Suspended, Complete, Cancelled
Constraint	Open, Assigned, In Resolution, Released, Waived, Cancelled, Closed
Delay Event	Open, Under Investigation, Impact Assessed, Mitigation Planned, Under Recovery, Resolved, Closed
Recovery Plan	Draft, Proposed, Active, Paused, Successful, Unsuccessful, Closed
Measurement Sheet	Draft, Submitted, Measured, Under Review, Returned, Agreed, Disputed, Certified, Rejected, Closed
Offline Mutation	Pending, Uploading, Synced, Conflict, Failed, Cancelled

36.1 Transition Requirements

- Every transition has a capability key
- Invalid transitions return allowed next states and reason
- Terminal records become read-only except controlled correction/closeout actions
- State timestamps and actors are immutable
- Bulk transitions validate every selected record and provide partial-failure results
- Imported states must pass the same domain rules as UI actions
- Platform Configuration may disable transitions but may not create invalid historical states

37. API Contract

37.1 Core REST Endpoints

```

GET    /work-progress/dashboard
GET    /projects/{projectId}/work-progress/dashboard
GET    /work-packages
POST   /work-packages
GET    /work-packages/{id}
PATCH /work-packages/{id}
POST   /work-packages/{id}/transition
POST   /work-packages/{id}/progress-updates
GET    /work-packages/{id}/progress-history
POST   /progress-updates/{id}/submit
POST   /progress-updates/{id}/verify
POST   /progress-updates/{id}/accept
POST   /progress-updates/{id}/reject
POST   /progress-updates/{id}/correct
GET    /projects/{projectId}/schedules
POST   /projects/{projectId}/schedules
POST   /schedules/{id}/calculate
POST   /schedules/{id}/activate-baseline
GET    /daily-site-reports
POST   /daily-site-reports
GET    /daily-site-reports/{id}
PATCH /daily-site-reports/{id}
POST   /daily-site-reports/{id}/submit
POST   /daily-site-reports/{id}/create-revision
GET    /constraints
POST   /constraints
PATCH /constraints/{id}
POST   /delays
POST   /recovery-plans
POST   /progress-evidence
GET    /measurement-sheets
POST   /measurement-sheets
POST   /measurement-sheets/{id}/certify
GET    /work-progress/reports
POST   /work-progress/reports/generate
GET    /work-progress/saved-views

```

```

POST /work-progress/saved-views
POST /work-progress/imports
GET  /work-progress/imports/{id}

```

37.2 API Standards

- OpenAPI contract and generated typed clients.
- Cursor pagination for large lists and timelines.
- Filter and sort allowlists; no arbitrary SQL-like query input.
- ETag or record_version required for mutable aggregate updates.
- Idempotency-Key required for offline, import, submit, accept, transition and certification mutations.
- Consistent error envelope with code, message, field errors, current version, allowed transitions and correlation ID.
- Bulk endpoints return per-record result and never hide partial failure.
- Long-running calculations/imports/reports return job IDs.
- Signed, time-limited upload/download URLs through Documents/media service.
- No direct cross-module database access from public APIs.

37.3 Example Progress Update Payload

```

{
  "idempotency_key": "mobile-uuid",
  "work_package_id": "uuid",
  "baseline_id": "uuid",
  "effective_at": "2026-08-05T14:00:00+02:00",
  "shift_id": "uuid",
  "location_id": "uuid",
  "method": "installed_quantity",
  "lines": [{
    "scope_line_id": "uuid",
    "period_quantity": "24.500",
    "unit": "m3",
    "evidence_ids": ["uuid"]
  }],
  "notes": "Concrete pour grid A-D",
  "record_version": 7
}

```

38. Backend Services, Events and Background Jobs

38.1 Domain Services

- Work Package Service
- Scope and Allocation Service
- Schedule and Critical Path Service
- Baseline Service
- Measurement and Verification Service
- Rollup Calculation Service
- Snapshot Service
- Daily Report Service
- Crew and Resource Log Service
- Constraint and Delay Service
- Recovery Scenario Service
- Evidence Service
- Subcontractor Measurement Service
- Certification Integration Service
- Forecast Service
- Data Quality Service
- Import/Export Service
- Search Indexing Service
- Offline Sync Service
- Audit Service

38.2 Domain Events

Event	Minimum payload / consumer examples
work_package.created	Tenant, company, project, package, actor; search and activity consumers.
work_package.status_changed	Old/new state, reason, effective time; Project, Tasks, Notifications later.
baseline.activated	Baseline version, scope, dates, checksum; snapshots and reports.
progress_update.submitted	Package, period, quantities, submitter; verification hook.
progress_update.accepted	Accepted lines, previous/new cumulative values; rollup, forecast, Project summary.
progress_update.corrected	Original, reversal/correction, reason; rollup and audit.
daily_report.submitted	Project/date/shift/revision; Documents/report generation.
constraint.created_or_due	Owner, required-by, blocking impact; Tasks/Notifications later.
forecast.changed	Old/new finish, reason and critical path; dashboard/calendar/notification hooks.
measurement_sheet.certified	Contract/period/quantities/value/source version; Finance and Contracts.
offline_sync.conflict	User/device/record/conflict type; user notification and observability.

38.3 Background Jobs

- Recalculate affected package/project rollups after accepted changes.
- Recompute schedule, float and critical path after schedule changes.
- Generate daily/weekly/monthly immutable progress snapshots.
- Recalculate forecasts and confidence factors.
- Detect stale packages, missing reports and overdue constraints.
- Process report and PDF/Excel generation.
- Generate thumbnails and media renditions.
- Validate evidence/document references.
- Rebuild search index.
- Process import batches and validation reports.
- Clean expired local/signed upload sessions without deleting accepted media.
- Retry outbox events and dead-letter failed integrations.
- Aggregate dashboard read models incrementally.

38.4 Event Reliability

- Transactional outbox
- At-least-once delivery
- Idempotent consumers
- Event schema version
- Correlation and causation IDs
- Retry with backoff
- Dead-letter monitoring
- Replay support
- No sensitive document content in event payloads

39. Frontend Architecture

39.1 Route and Feature Structure

```
work-progress/
  dashboard/
  work-packages/
  work-package-page/
  schedule/
  lookahead/
  daily-reports/
  field-capture/
  constraints-delays/
  evidence/
```

```

subcontractor-progress/
measurement-certification/
reports/
settings/
shared/calculation-drilldown/
shared/offline-sync/

```

39.2 Major Components

- ProgressKpiStrip
- ProgressCurveChart
- CalculationDrilldown
- WorkPackageTable
- WorkPackageBoard
- LocationProgressMatrix
- ScheduleGantt
- BaselineComparison
- LookaheadBoard
- DailyReportEditor
- QuickProgressCapture
- QuantityMeasurementGrid
- CrewLogEditor
- EquipmentLogEditor
- ConstraintDrawer
- DelayImpactPanel
- RecoveryScenarioPanel
- EvidenceGallery
- PhotoAnnotator
- MeasurementSheetGrid
- CertificationComparison
- DataQualityPanel
- SyncQueue
- ConflictResolver
- ActivityTimeline

39.3 State Management

- Server state through the platform query/cache standard.
- Route-driven filters where shareable.
- Local form drafts with explicit dirty state.
- Normalized entities for large schedule/package views.
- Optimistic UI only for reversible personal actions and drafts; official acceptances wait for server confirmation.
- Virtualized rows and memoized chart data.
- Web worker for heavy client-side preview/validation where safe.
- Offline store isolated by tenant/project and cleared on logout/access removal.
- Feature flags and Platform Configuration loaded centrally.

39.4 Error and Empty States

- No baseline configured
- No work packages
- No progress updates
- Insufficient data for forecast
- Calculation in progress
- Stale read model
- Permission denied
- Record changed by another user
- Offline with unsynced data
- Partial bulk failure
- Document/evidence unavailable
- Source module temporarily unavailable

WP-088

Official progress must not update optimistically before the server accepts and recalculates the affected aggregate.

WP-089

Every calculation or integration error must preserve the user draft and provide a retry or recovery path.

40. Offline Synchronization and Conflict Resolution

Conflict type	Resolution rule
Append-only photo/evidence	Auto-merge if unique checksum/client ID; duplicate warning otherwise.
Independent daily report lines	Merge when different line IDs and report remains editable.
Same scalar field edited	Present server and local values; user chooses or creates a revision.
Status changed on server	Reject invalid local transition and show current allowed actions.
Scope quantity changed	Revalidate local progress against new scope; do not auto-truncate.
Report locked on server	Local edits become a proposed correction/revision, not overwrite.
Progress already accepted	Duplicate idempotency returns original result; conflicting quantity requires correction flow.
Access removed	Stop sync, retain encrypted local draft temporarily according to policy and allow secure export only if authorized.
Deleted/archived reference	Prevent submission and request relink or cancellation.
Attachment upload failed	Keep record draft and retry attachment independently; submission rules depend on mandatory evidence.

40.1 Conflict UX

- Human-readable conflict reason
- Server version and local version
- Affected fields/lines highlighted
- Recommended resolution
- Keep server / use local / merge / create revision / discard draft actions as valid
- No raw JSON required for normal users
- Audit of chosen resolution
- Bulk conflict review for field supervisors

WP-090	The system must never resolve an accepted-quantity conflict by last-write-wins.
WP-091	Offline drafts must be encrypted at rest using the platform standard and removed when retention/access policy requires it.

41. Validation, Concurrency, Security and Audit

41.1 Validation Rules

- Unique work package code within configured project/company scope.
- Mandatory project, accountable owner, measurement method and baseline before Planned/Ready state.
- Valid unit of measure and conversion.
- Non-negative quantities, weights, durations and hours unless using explicit reversal records.
- Cumulative accepted quantity within approved scope or authorized overrun state.
- Weight totals within configured tolerance.
- No schedule dependency cycles.
- Actual finish cannot precede actual start.
- Forecast dates respect calendar/dependency calculation unless authorized override is recorded.
- Daily report labor/equipment totals reconcile within tolerance.
- No duplicate report scope/date/shift unless explicitly allowed.
- No duplicate claim line for same contract/BOQ/location/period without revision relationship.
- Required evidence and quality/HSE gates satisfied before configured acceptance/completion state.
- All source record references belong to the same tenant and compatible project/company scope.
- Archive/close prohibited while mandatory unresolved records remain.

41.2 Concurrency

- Optimistic concurrency using record_version/ETag

- Atomic accepted progress update plus cumulative quantity update
- Database constraints for unique idempotency keys and active baseline
- Serializable or locked transaction for certification cumulative quantities
- Asynchronous rollup with versioned read model and pending-calculation indicator
- No distributed transaction across modules; use outbox/saga and reconciliation

41.3 Capability Keys

Capability	Purpose
work_progress.view	View authorized progress records.
work_progress.manage_packages	Create/edit/archive work packages.
work_progress.manage_schedule	Edit schedule, dependencies and working versions.
work_progress.activate_baseline	Activate baseline through configured control.
work_progress.submit_progress	Submit field progress.
work_progress.verify_progress	Verify/accept/reject progress.
work_progress.correct_progress	Create corrections/reversals.
work_progress.manage_daily_reports	Create/edit/submit reports.
work_progress.lock_daily_reports	Verify/lock or issue revisions.
work_progress.manage_constraints	Create/update/release constraints.
work_progress.manage_delays	Create/assess/close delays.
work_progress.manage_recovery	Create/promote recovery plans.
work_progress.view_commercial	View claim/certification/cost context.
work_progress.manage_measurement	Measure/review claim lines.
work_progress.certify	Certify progress through configured control.
work_progress.override_gate	Apply authorized readiness/acceptance override.
work_progress.export	Generate/export reports.
work_progress.configure	Manage module settings and methods.

41.4 Security

- Tenant isolation on every query and mutation.
- Company/project scope enforcement.
- Current phase may configure broad authenticated access, but backend authorization calls are mandatory.
- Field-level protection for commercial, workforce, client and confidential evidence data.
- Signed short-lived media URLs and malware scanning.
- Rate limits and upload limits.
- No secrets or personal salary/health data in progress records.
- Access revocation invalidates offline tokens and media URLs.
- Export watermark/classification according to policy.
- Audit access to sensitive commercial reports where configured.

41.5 Audit and Immutability

- Actor, organization, role snapshot and device/session
- Before/after values for mutable fields
- State transition and reason
- Baseline/calculation version
- Effective and transaction timestamps
- Source/correlation/causation IDs
- Export/import events
- Permission/override use
- Offline conflict resolution
- Accepted corrections/reversals
- Document/evidence link changes
- Audit retention and legal hold

WP-092	Accepted progress, locked daily reports, active baseline snapshots and issued certificates must be immutable business records.
WP-093	Audit history must be searchable and exportable without allowing audit events to be edited or deleted through normal application APIs.

42. Performance, Scalability and Observability

42.1 Scale Targets

Area	Target / strategy
Projects and packages	Support portfolio views with thousands of projects and individual projects with tens of thousands of packages/activities.
Progress lines	Millions of append-only measurement lines through partitioning/index strategy and cursor pagination.
Daily reports	Multi-year retention with project/date indexes.
Media	Object storage/CDN via Documents/media; thumbnails and lazy loading.
Dashboards	Precomputed/incremental read models; no full raw-table aggregation on every request.
Schedule calculation	Asynchronous job for large schedules with progress/status feedback.
Search	Dedicated indexed search with tenant/project security filters.
Exports	Asynchronous generation and expiring download links.
Offline sync	Batch mutation endpoint and attachment concurrency controls.

42.2 Performance Budgets

- Typical dashboard first meaningful data within platform performance budget
- Table/filter response optimized for interactive use
- Quick capture opens from cached project data
- No full-resolution media in list views
- Incremental chart loading
- Background recalculation clearly indicated
- Batch endpoints to avoid chatty field synchronization
- Database query plans and index monitoring

42.3 Observability

- Structured logs with tenant-safe correlation IDs
- Metrics for API latency/error, job duration/failure, schedule calculation, rollup lag, sync conflict and report generation
- Distributed traces across module events
- Dead-letter and outbox lag alerts
- Data-quality and reconciliation dashboards
- Audit of failed authorization and invalid transitions
- Synthetic monitoring of critical field and dashboard flows
- No sensitive evidence content in logs

WP-094	Dashboard read models must expose their source calculation time and may show Calculating when not yet synchronized; they must not silently show stale values as current.
WP-095	The platform must support reconciliation jobs that compare accepted progress lines, cumulative quantities, rollups and snapshots and report any inconsistency.

43. Testing Strategy and Quality Gates

43.1 Unit Tests

- Measurement formulas and rounding
- Weight/rollup validation
- Quantity caps and reversals
- Lifecycle transitions
- Schedule dependency and calendar calculations
- Baseline immutability
- Forecast rate calculation
- Data quality score factors
- Status classification
- Authorization policy adapters
- Idempotency behavior

43.2 Integration Tests

- Project hierarchy references
- Documents revision/evidence links
- Inventory issue and receipt links
- Asset usage links
- Finance actual cost read and certification event
- Contract/BOQ/milestone links
- Procurement delivery constraints
- Tasks creation without progress side effects
- Outbox/event replay
- Search indexing
- Report generation
- Offline batch sync

43.3 End-to-End Tests

- Create package -> baseline -> Ready -> field update -> verification -> rollup -> dashboard.
- Daily report offline -> photos -> reconnect -> submit -> lock -> revision.
- Constraint blocks package -> task/removal -> release -> readiness update.
- Contractor claim -> measurement -> accepted progress -> certification event.
- Rebaseline without changing historical original-baseline report.
- Correction/reversal after accepted wrong quantity.
- Concurrent users submit overlapping quantity and one is rejected safely.
- Large project table, schedule and report performance.
- Access removed while offline.
- Platform Configuration disables a feature without dead links.

43.4 Calculation Golden Datasets

QA must maintain deterministic datasets covering quantity, milestone, composite, rebaseline, cancellation, variation, reversal, schedule, forecast and earned-value cases. Expected inputs and outputs must be version-controlled and rerun whenever the calculation engine changes.

43.5 Non-Functional Testing

- Load and stress
- Long-running schedule/import/report jobs
- Offline interruption and retry
- Security/tenant isolation
- Media upload malware/size limits
- Accessibility
- Responsive browsers and 320 px width
- Localization/timezone/DST
- Backup/restore and event replay
- Data migration reconciliation

WP-096

A calculation-engine release cannot be deployed if golden datasets change unexpectedly or historical snapshot reproduction fails.

WP-097

Tenant-isolation tests must cover lists, dashboards, search, exports, media, events and cached/offline data.

44. Import, Migration and Rollout

44.1 Supported Imports

- Work package register
- WBS and location mappings
- BOQ/quantity allocations
- Schedule activities, dependencies and calendars
- Baseline dates and weights
- Historical progress by period
- Daily reports
- Contractor claims and measurement sheets
- Constraints/delays
- Crew/equipment logs
- Photo/evidence manifest
- External IDs from spreadsheets or planning tools

44.2 Import Flow

1. Upload file or select source
2. Choose tenant/company/project and import type
3. Map columns/entities/units/timezone
4. Validate references and duplicates
5. Preview hierarchy, quantities, weights and dates
6. Resolve blocking errors
7. Run dry-run calculations
8. Confirm import
9. Create records and immutable import report
10. Reconcile totals and expose rollback/correction strategy

44.3 Migration Rules

- Preserve source external IDs and file checksum.
- Do not infer official acceptance from an unlabeled spreadsheet percentage.
- Historical progress must be classified as submitted, accepted or unknown; unknown values remain clearly labeled.
- Imported baselines require version and effective-date treatment.
- Unit conversions are explicit and logged.
- Duplicate detection uses project/package/date/location/source keys.
- Large imports are resumable and asynchronous.
- Rollback uses import-batch archive/reversal rules, not destructive deletion after records are referenced.

WP-098

The importer must not silently normalize weights or quantities to make invalid data pass. It must show discrepancies and require a controlled choice.

45. Implementation Phases

Phase	Scope	Exit criteria
Phase 1 - Foundations	Project configuration, work packages, hierarchies, statuses, measurement methods, basic baseline, progress updates and audit.	Accepted quantity progress produces correct package/project rollups and drill-down.
Phase 2 - Field Operations	Daily reports, crews, equipment, evidence, mobile-responsive capture and offline queue.	One full site day can be recorded offline and synchronized without duplication.
Phase 3 - Schedule and Control	Activities, dependencies, calendars, baselines, lookahead, constraints, delays and forecast.	Planned/actual/forecast and critical-path exceptions are reliable.
Phase 4 - Resource Integrations	Inventory, Procurement, Assets, Teams/HR and Documents integration.	Materials/equipment/workforce context is source-linked and reconciled.
Phase 5 - Commercial Measurement	Subcontractor submissions, measurement sheets, claims, certification hooks and Finance/Contracts events.	Claimed/measured/accepted/certified states are independently auditable.

Phase	Scope	Exit criteria
Phase 6 - Analytics and Hardening	Reports, data quality, portfolio views, performance, accessibility, security, import and observability.	Scale, QA, security and acceptance gates pass.
Later Platform Integration	QA/QC, HSE, Calendar, Notifications, BIM/3D, portals and final Approvals/Workflow Engine.	Integrated through stable contracts without rewriting module ownership.

45.1 Rollout Strategy

- Pilot on one active project and one completed project for historical comparison
- Run parallel reporting for a defined period
- Reconcile package totals, baseline, daily reports and certificates
- Train site roles through scenario-based workflows
- Measure reporting timeliness and correction rate
- Expand by project template and discipline
- Do not retire existing records until export/retention and dispute requirements are satisfied

46. Acceptance Criteria

ID	Acceptance criterion
AC-01	Projects can create and manage work packages with WBS, location, BOQ, schedule, contractor and document links.
AC-02	Tasks and Work Progress remain separate records and task completion cannot directly increase official physical progress.
AC-03	Every official progress value identifies accepted/submitted/claimed/certified/paid type, baseline and effective date.
AC-04	At least quantity, completed-unit, weighted-milestone, activity-weighted, manual and composite measurement methods are supported.
AC-05	Only accepted progress contributes to official rollups.
AC-06	Accepted quantities cannot exceed approved scope without an explicit overrun/change state.
AC-07	Duplicate quantity/location/unit contributions are detected.
AC-08	Accepted corrections use reversal/correction records and preserve history.
AC-09	Original and current baselines remain available after rebaseline.
AC-10	Historical snapshots are reproducible with calculation version and input IDs.
AC-11	Schedules support calendars, milestones, FS/SS/FF/SF dependencies, lag, float and critical/near-critical state.
AC-12	Dependency cycles are rejected with a useful error.
AC-13	Project dashboards show planned, accepted actual and forecast progress with drill-down.
AC-14	High overall progress cannot hide late critical packages; critical delay remains visible.
AC-15	Daily reports capture weather, labor, equipment, work, deliveries, issues, evidence and next plan.
AC-16	Submitted/locked daily reports are immutable and corrected by revision.
AC-17	Daily report PDF is generated and linked without duplicating the source record.
AC-18	Mobile users can capture progress, daily data, constraints and photos at 320 px width.
AC-19	Offline records use idempotent synchronization and display unsynced/conflict states.
AC-20	Accepted-progress conflicts are never resolved by last-write-wins.
AC-21	Crew and equipment productivity calculations preserve source hours and context.
AC-22	Inventory stock is never changed directly by Work Progress.
AC-23	Asset identity and meter/maintenance lifecycle remain authoritative in Assets.
AC-24	Material shortages and equipment unavailability can block readiness and affect forecast.
AC-25	Contractor reported, measured, accepted, claimed, certified and paid values are separate.
AC-26	Measurement sheets show reasons for claimed-to-measured-to-certified differences.
AC-27	Constraints have owner, required-by date, affected scope, impact and release evidence.
AC-28	Delay records separate observed facts from alleged/formal responsibility.
AC-29	Recovery plans quantify schedule, resource and cost assumptions.
AC-30	Progress photos support timestamp, package, location, annotation, before/after and confidentiality.
AC-31	Evidence links retain exact Document revision IDs where required.
AC-32	QA/QC and HSE gates can block package readiness without duplicating those modules.
AC-33	Pending variation scope remains separate from approved progress denominator.

ID	Acceptance criterion
AC-34	Forecast returns Insufficient Data when inputs are inadequate.
AC-35	Data quality and confidence indicators expose contributing factors.
AC-36	Reports disclose baseline, effective date, filters, generator and progress type.
AC-37	Saved views preserve layout, filters, grouping, columns and baseline per user.
AC-38	All cross-module writes use APIs/events and correlation IDs, not direct database writes.
AC-39	All mutations validate tenant/company/project scope and common authorization hooks.
AC-40	Issued records and audit events cannot be edited or deleted through normal APIs.
AC-41	Large lists use cursor pagination/virtualization and remain performant.
AC-42	Large schedule calculation, imports and reports are asynchronous with progress state.
AC-43	Dashboard read models disclose calculation time and stale/calculating state.
AC-44	Golden calculation datasets pass before release.
AC-45	Tenant isolation covers dashboards, search, exports, media and offline cache.
AC-46	Platform Configuration can disable features without dead links or blank spaces.
AC-47	Premium motion is restrained, accessible and does not delay field workflows.
AC-48	Charts have accessible text/table alternatives and status is not color-only.
AC-49	Import validates units, weights, hierarchy, duplicate progress and baseline identity.
AC-50	An auditor can reconstruct a project progress value from dashboard to accepted measurements and evidence.

47. Page and Flow Registry

- Global Work Progress Dashboard
- Project Work Progress Dashboard
- Work Packages Page
- Create/Edit Work Package
- Work Package Page
- Quick Progress Capture
- Progress Update Page/Drawer
- Progress Verification Queue
- Progress Calculation Drill-down
- Construction Schedule
- Schedule Activity Editor
- Baseline Manager
- Baseline Comparison
- Lookahead Planning
- Lookahead Publication
- Readiness Matrix
- Daily Site Reports Page
- Daily Site Report Editor
- Daily Report Revision Flow
- Crew and Labor Log
- Equipment Usage Log
- Material/Delivery Log
- Constraints Register
- Constraint Page/Drawer
- Delay Register
- Delay Page
- Recovery Plan Page
- Recovery Scenario Comparison
- Progress Photos Gallery
- Photo Capture and Annotation

- Evidence Coverage View
- Subcontractor Progress Page
- Contractor Submission Flow
- Measurement Sheets Page
- Measurement Sheet Page
- Certification Flow Hook
- Data Quality Dashboard
- Progress Reports
- Report Generation Job
- Saved Views
- Import Wizard
- Import Validation Report
- Offline Sync Queue
- Conflict Resolution
- Work Progress Settings
- Platform Configuration Registry Entries
- Activity and Audit Timeline

48. Future Roadmap

- QA/QC inspections, ITPs, NCR, snagging, handover and warranty integration.
- HSE permits, incidents, risk assessments, safety observations and stop-work integration.
- Calendar, meetings, notifications and final Approvals/Workflow Engine.
- BIM/IFC and 3D element-based progress visualization.
- 4D schedule simulation and model colorization by planned/actual/quality state.
- Drone/photo comparison and geospatial progress capture.
- External Primavera P6 and Microsoft Project bidirectional integration.
- Weather service and site sensor integration.
- Native mobile apps where browser capability is insufficient.
- Client and subcontractor portals with controlled record-level access.
- AI-assisted photo classification, quantity suggestion, delay-risk detection and narrative summaries only after reliable source data exists.
- Predictive production and resource optimization with explainable assumptions.
- Digital twin and IoT equipment/asset telemetry.

Roadmap rule

Automation and AI must never be used to hide weak source data. Future intelligence features may suggest, classify or forecast, but official progress still requires controlled measurement, verification and audit.

49. Final Architectural Rule

The Work Progress Module must not become another percentage dashboard or digital version of an unreliable spreadsheet. It must be the complete, evidence-backed and historically reproducible operating record of physical construction execution.

Permanent platform principle

One approved scope, one identified baseline, one accepted physical-progress history, and full traceability from project rollup to the exact measured work, location, date, party and evidence. Plans may change, forecasts may change and corrections may occur, but history must never be rewritten.

- Projects provide context.
- Work packages define executable scope.
- Baselines define the plan.
- Accepted measurements define physical progress.
- Daily reports explain site reality.
- Constraints and delays explain lost time.
- Resources explain productivity.
- Evidence supports the record.
- Contracts and Finance govern commercial consequences.
- Audit preserves accountability.